Program 6: Vacuum Cleaner Problem

Aim

To develop a Python program to solve the Vacuum Cleaner Problem by cleaning dirty rooms.

Algorithm

1. Initialize the rooms with clean or dirty states.
2. Place the vacuum cleaner in the initial location.
3. Check whether the current room is dirty.
4. If dirty, clean the room.
5. Move the vacuum cleaner to the next room.
6. Repeat until all rooms are clean.
7. Display the final state of the rooms.

```python
def vacuum_cleaner():
    room = {
        'A': 'Dirty',
        'B': 'Dirty'
    }

    cost = 0
    location = input("Enter the initial location (A/B): ").upper()

    if location not in room:
        print("Invalid location!")
        return

    print("\nInitial Room Status:")
    for place, status in room.items():
        print("Room", place, ":", status)

    while True:
        print("\nVacuum Cleaner is in Room", location)

        if room[location] == 'Dirty':
            print("Room", location, "is dirty.")
            print("Vacuum cleaner is cleaning Room", location)
            room[location] = 'Clean'
            cost += 1
        else:
            print("Room", location, "is already clean.")

        print("Current Room Status:", room)

        if all(status == 'Clean' for status in room.values()):
            print("\nBoth rooms are clean.")
            break

        if location == 'A':
            location = 'B'
            cost += 1
        else:
            location = 'A'
            cost += 1
```

File Edit Shell Debug Options Window Help

```
Python 3.13.0 (tags/v3.13.0:60403a5, Oct  7 2024, 09:38:07) [MSC v.1941 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/ashik/AppData/Local/Programs/Python/Python313/program 6.py =
Enter the initial location (A/B): A

Initial Room Status:
Room A : Dirty
Room B : Dirty

Vacuum Cleaner is in Room A
Room A is dirty.
Vacuum cleaner is cleaning Room A
Current Room Status: {'A': 'Clean', 'B': 'Dirty'}

Vacuum Cleaner is in Room B
Room B is dirty.
Vacuum cleaner is cleaning Room B
Current Room Status: {'A': 'Clean', 'B': 'Clean'}

Both rooms are clean.

Final Room Status:
Room A : Clean
Room B : Clean
Total Cost: 3
Cleaning completed successfully!
```

Result

Thus, the Vacuum Cleaner Problem was implemented successfully using Python, and the dirty rooms were cleaned.